

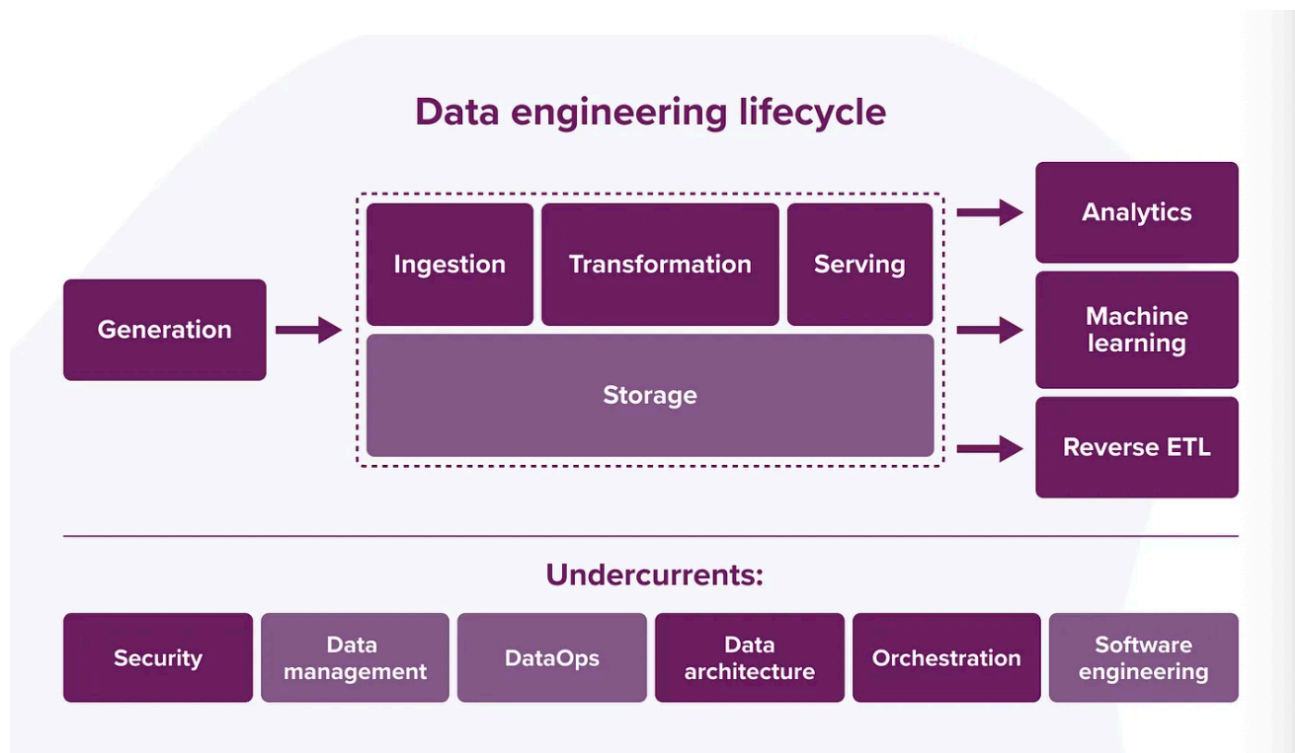
Data Engineering Lifecycle

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This lifecycle covers the entire process of transforming raw data into a useful end product. It involves several stages, each for a particular task and has a specific responsibility. This ensures data is handled efficiently and effectively, from its generation, to final consumption.

Overview:

- Generation
- Storage
- Ingestion
- Transformation
- Serving Data



Stage 1 - Generation

Source Systems are the origin points of data. This means that they are the places from where the data which the system needs is generated. These can range from SAPs to IoT devices to transactional databases. Data Engineers interact with these systems to collect raw, unfiltered data.

To master generation, data engineers must be familiar with these source systems and the data being generated from them. They must consider factors such as, the format of the

data being generated, the persistence of the data (should it be long term or short term), how frequently the data is generated, how many duplicates are generated at regular intervals, etc. and design systems accordingly.

Stage 2 - Storage

The choice of storage solution can impact the efficiency of data processing, transformation and serving. When selecting a storage solution, some factors engineers have to keep in mind are

- Ensure the storage solution supports the required operation speeds for the architecture.
- Check whether the storage system can create bottlenecks for processes which occur after the storage of data.
- Consider scalability of the storage solution, depending upon the data needs.
- Check whether data retrieval adhere to service-level agreements and whether the storage solution provides support for complex queries or merely storage.

Data Access Frequency is one of such factors which engineers have to keep in mind when designing systems. Data can be "Hot" which means its frequently accessed, needs fast retrieval, and is stored in high speed storage solutions. IT can also be "Lukewarm" which means its accessed occasionally, and is stored in moderately fast storage solutions. Finally it can be "Cold" which means its rarely accessed, thus can be stored in archival storage solutions.

Some examples of storage solutions are Data Warehouses, Lakehouses, Data Lakes.

Stage 3 - Ingestion

Data Ingestion is the process of gathering and moving data from various source systems into a centralized *Data Repository* . This stage is required for ensuring data is available for further processing and analysis.

Considerations to be made in this stage are

- Identify the purpose of the ingested data and how it can be potentially be reused.
- Ensure that the source systems and ingestion processes are reliable and data is available when needed.
- Determine where the data will be stored and how it will be accessed.
- Understand how often the data will be accessed.
- Assess the volume and format of the incoming data, and what potential transformations are required.

Types of Ingestion

- **Batch Ingestion** - Processes large chunks / quantity of data at scheduled intervals. Used when real time processing is not required.
- **Streaming Ingestion** - Provides data to systems in near real time scenarios. Most used for scenarios which require immediate data availability.

Push Model - In this model, the source systems write data directly to the target system. Useful when immediate data transfer is required.

Pull Model - In this model, the target system retrieves data from source systems. Most suitable for periodic data collection.

Stage 4 - Transformation

This stage involves converting raw data into a format suitable for analysis and consumption. This adds value to data and making it useful for systems which make use of this data.

Considerations to be made

- Assess the cost and ROI for the transformation processes.
- Ensure transformations are as simple as possible.
- Incorporate business logic to ensure data transformations support organisational needs.

Transformation Methods

- **Data Cleaning** - Correcting or Removing inaccurate records.
- **Schema Transformation** - Changing the data schema to a more usable structure.
- **Normalization** - Organising data to reduce redundancy. For example, breaking down a large customer table into separate tables for customers, orders, and products.
- **Aggregation** - Summarizing data for reporting purposes. Eg - Calculate total sales of a product for this month.

Stage 5 - Serving Data

Serving data means providing processed data to end-users for decision making, operational purposes and advanced analytics. This is where the value of data engineering shines.

Further Reading

- <https://www.marktechpost.com/2023/02/27/the-concept-of-data-generation/>
- <https://www.ibm.com/think/topics/data-storage>
- <https://www.qlik.com/us/data-ingestion>

- <https://www.ibm.com/think/topics/data-ingestion>